Web Programming - Assignment 2

Instructor: Mahhek Tahir

Event Management System using React & Redux Toolkit

Objective

Develop a React-based Event Management System that allows users to explore events, view event details, register for events, and manage their registered events.

The application must use:

- React.js
- Redux Toolkit
- React Router
- Ticketmaster Discovery API

The assignment is intentionally open-ended. Students are expected to explore the technologies and documentation and make appropriate implementation and design decisions.

1. User Authentication

- Provide User Registration, Login, and Logout.
- Maintain user-specific authentication/session state.
- The registration form should collect at least Name, Email, and Password.
- Manage authentication state using Redux Toolkit.
- Use Local Storage so that the login state can persist when the page is refreshed.

Note: This is a frontend authentication simulation. A real backend authentication system is not required.

2. Event Exploration

- View available events.
- Search for events.
- Filter events using relevant criteria.
- Explore upcoming events.
- View event categories.
- View individual event details.

Event information must be obtained from the Ticketmaster Discovery API.

3. Event Details

- Event name
- Date and time
- Venue
- Location
- Category/type
- Event image
- Other useful information available through the API

4. Event Registration

- A logged-in user should be able to register for an event.
- Prevent duplicate registration.
- Provide appropriate feedback after registration.
- Allow a user to cancel their registration.

- Associate registration information with the logged-in user.

5. My Events

- Provide a My Events section where users can view their registered events.
- Distinguish between upcoming and completed events.
- Allow users to cancel a registration where appropriate.

6. Redux Toolkit

Redux Toolkit must be used to manage the application's major shared state. Students should determine an appropriate Redux structure for their application.

The application should demonstrate meaningful use of:

- configureStore
- createSlice
- Actions
- Reducers
- useSelector
- useDispatch

Possible areas for Redux state include:

```
Authentication | Events | Registrations
```

7. React Router

Use React Router to provide appropriate navigation between pages. Relevant views may include:

- Login
- Register
- Events
- Event Details
- My Events

Students should decide the appropriate route structure.

8. API Requirement

Use the Ticketmaster Discovery API to fetch event information. The Discovery API supports searching and retrieving event, attraction, venue, and classification information. Students should explore the documentation and determine the appropriate endpoints and parameters.

Get Your API Key

Register and obtain your API key from:

https://developer.ticketmaster.com/

API Documentation

https://developer.ticketmaster.com/products-and-docs/apis/discovery-api/v2/

Example API URL:

```
https://app.ticketmaster.com/discovery/v2/events.json?apikey=YOUR_API_KEY
```

> Important: Do not commit your API key or .env file containing the key to GitHub. Use environment variables for local development.

9. Git & Version Control

Git must be used throughout the development of this assignment.

- Create a GitHub repository for the project.

- Make multiple meaningful commits throughout development.
- Use clear and descriptive commit messages.
- Commit features and improvements regularly.
- Do not complete the entire project and submit it as a single commit.

Example commit messages:

```
Initial React project setup
Add authentication pages
Implement Redux authentication
Add Ticketmaster API integration
Add event listing
Implement event search
Add event registration
Add My Events
Add protected routes
Improve responsive UI
```

Pull Requests

You are encouraged to use Pull Requests (PRs) as part of your development workflow. For example, develop a feature in a separate branch and create a Pull Request to merge it into the main branch.

For this assignment, you may review and approve your own Pull Requests. The purpose is to gain practical experience with branches, commits, Pull Requests, and GitHub-based development workflows.

> Your repository should demonstrate actual development history, rather than simply containing the final project.

10. User Interface

- Responsive
- User-friendly
- Consistent in design
- Properly organized into reusable React components

Students are free to design the interface according to their own ideas.

11. GitHub Submission

The complete project must be uploaded to GitHub.

The repository should contain:

- Complete source code
- package.json
- README with setup and execution instructions
- .gitignore
- Meaningful commit history

> Submission: Submit only your GitHub repository URL.

Example:

```
https://github.com/yourusername/event-management-system
```

Make sure the repository is accessible to the instructor.

12. Important

This assignment is intentionally open-ended. You are expected to:

- Explore the Ticketmaster API documentation.
- Decide which API features are useful.

- Design your own UI.
- Decide how to structure your Redux store.
- Create reusable React components.
- Handle loading and error states.
- Implement appropriate form validation.
- Make reasonable implementation decisions where requirements are not explicitly specified.

Copying a tutorial or implementing only the minimum required screens will not be considered sufficient.

API details verified against Ticketmaster's current Discovery API documentation. The API uses the apikey query parameter, and the developer portal identifies the Consumer Key as the API key. The current default quota is 5,000 requests/day with a 2 requests/second limit.